

OVERVIEW OF HARDWARE COMPONENTS

1. INTRODUCTION TO COMPUTER HARDWARE



WHAT IS HARDWARE?

Physical parts of a computer that can be seen and touched.

Examples:



Monitor



Keyboard



Mouse



Motherboard



CPU



RAM



Hard Disk



Hard Disk



HARDWARE vs SOFTWARE

Physical components

Programs & applications

Physical components

Programs & applications

| Can be touched

| Cannot be touched

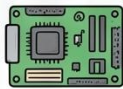
Example: Keyboard

Example: Microsoft Word

QUESTION: Can a computer work with only hardware and no software?

ANSWER: NO. Hardware requires software instructions to perform tasks.

2. MOTHERBOARD



Main circuit board. Connects all hardware and allows them to communicate.

Functions

Connects CPU, RAM, storage, expansion cards

Distributes power

Enables communication between devices

Enables communication
RAM, Slots
PCIe Slots
Storage cat
Power Connectors

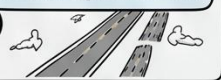


Main parts on motherboard

1. CPU Socket
2. RAM Slots
3. PCIe Slots
4. Storage Connectors (SATA, M.2)
5. Power Connectors

Simple Analogy:

Think of it as a city road network. All vehicles (data) travel through these roads.



Laptop



Keyboard



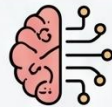
Mouse



Hard Disk

3. CPU (CENTRAL PROCESSING UNIT)

Defined: The brain of the computer.



Responsibilities

Processes instructions

Performs calculations

Controls operations

CPU components

1. Arithmetic Logic Unit (ALU)

2. Control Unit (CU)

3. Cache Memory

4. Cache Memory
(x2 Arithmetic Logic Unit for ALU)



Popular CPU Manufacturers:

Intel, AMD, Green

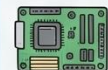
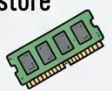
When you open a browser:

1. Chrome →
2. Open:
3. Compute →
3. Compute →
4. Brew:

4. RAM (RANDOM ACCESS MEMORY)



Temporary memory used to store data currently being used.



Characteristics

Fast access speed

Temporary storage

Data is lost when power off

Why RAM is Important:

More RAM allows...

- Better multitasking
- Faster application performance
- Smoother user experience



Example programs:

- Chrome
- Word
- Spotify

Analogy

RAM is like your study desk.
Larger desk = More books open.
Smaller desk = Less workspace.

Why RAM is important:

More RAM allows...

- Better speed
- Faster app performance
- Smoother user experience



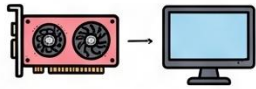
GPU, STORAGE, & SYSTEM FLOW GUIDE



(Day 3 mention is removed)

5. GPU (Graphics Processing Unit)

WHAT IS A GPU?



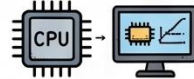
GPU is responsible for processing graphics & displaying images on the monitor.

USES:

- Gaming
- Video Editing
- 3D Modeling
- Artificial Intelligence
- Machine Learning (Algorithm flowchart)

TYPES OF GPU:

INTEGRATED GPU



Built inside the CPU.
Example: Intel UHD Graphics

DEDICATED GPU



Separate graphics card.
Examples: NVIDIA GeForce, AMD Radeon

ANALOGY:



CPU = GENERAL MANAGER

GPU = GRAPHICS SPECIALIST



6. Storage Devices

STORAGE DEVICES: Save data permanently.



HDD (Hard Disk Drive)

Features:

- Uses spinning magnetic disks
- Larger storage capacity
- Lower cost.

Advantages:

- Affordable
- High capacity.

Disadvantages: Slower speed

- Mechanical parts wear out.



SSD (Solid State Drive)

Features:

- No moving parts
- Uses flash memory
- Faster performance.

Advantages: Fast boot

- Fast file access
- Durable.

Disadvantages: More expensive than HDD.

	HDD	vs	SSD
Speed	Slow		Fast
Moving Parts	Yes		No
Durability	Lower		Higher
Cost	Cheaper		More Expensive
Boot Time	Longer		Shorter

How All Components Work Together

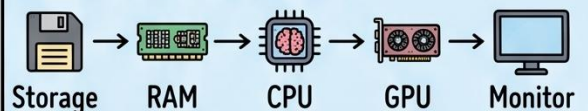
How All Components Work Together

FLOW OF SYSTEM ACTION:

Example: Opening Microsoft Word

- User clicks Word icon.
- SSD/HDD sends program data.
- RAM temporarily stores active data.
- CPU processes instructions.
- GPU displays application.
- Motherboard connects everything.

SYSTEM FLOW SUMMARY:



SUMMARY

TODAY WE LEARNED:

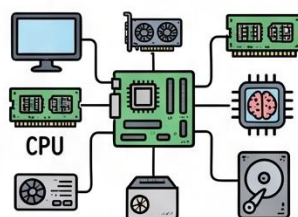
- ✓ Motherboard (Connects)
- ✓ CPU (Brain)
- ✓ RAM (Temp memory)
- ✓ GPU (Graphics)
- ✓ HDD (Traditional storage)
- ✓ SSD (Fast modern storage)

HOMEWORK QUESTIONS

- Function of motherboard
- Brain of computer
- Difference RAM/Storage
- SSD advantages
- Role of GPU

ASSIGNMENT

DRAW BASIC PC HARDWARE DIAGRAM



Sample: _____

Motherboard, =SSD/HDD

(Power Supply, etc.)

PRACTICAL HARDWARE EXPLORATION

COMPUTER DETECTIVE ACTIVITY:

Task 1: Know Your Computer

- Press **Win** + **R**, type cmd, and press Enter.
 - Type: hostname (Display computer name)
 - Type: whoami (Display user account)
 - Type: systeminfo (Detailed system info)
- What is the name of the computer? _____
- Which version of Windows is installed? _____

Task 2: Explore Hardware Components

- Type: wmic cpu get name (Windows Management Instrumentation Command-line) (displays the processor model).



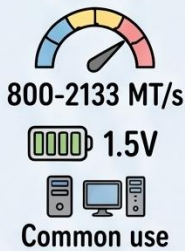
RAM AND STORAGE



TYPES OF RAM

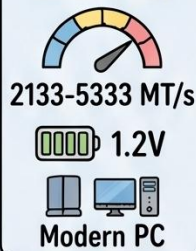
DDR3

Old PCs



DDR4

Modern PCs



DDR5

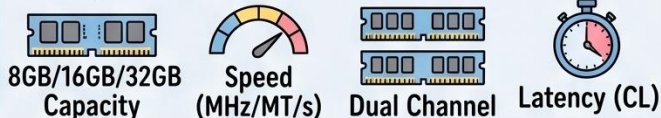
High-Performance



- DDR4: faster and efficient
- DDR5: higher bandwidth
- Generational notch compatibility



- Important terms:

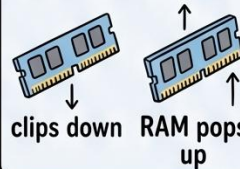


SAFETY FIRST

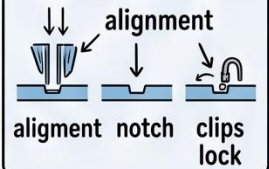
- ⏻ shut down PC (off button)
- 🔌 unplug (power cord unplug)
- ⚡ discharge (lightfinge)
- 👉 touch metal (hand grounding)

Practical Activity

Removing RAM

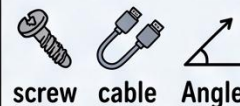


Reinstalling RAM

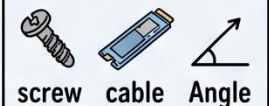


Check: ☒ ☒ ☒ ☒

SATA Drive Removal/Installation



M.2 NVMe SSD Removal/Installation



Storage Types

HDD

(HARD DISK DRIVE)

spinning disks and magnets

- advantages

📦 Large capacity

💰 Cheap coin

- disadvantages

🐢 Slow turtle

🔊 Noisy broken

SSD

(SOLID STATE DRIVE)

flash memory chips

- advantages

🚀 fast rocket

🛡️ reliable shield

🔊 silent speaker

- disadvantages

💰 expensive coin

	HDD	SATA SSD	NVMe SSD
Cost	Cheapest	Moderate cost	Higher cost
Speed	Slow	Fast	Very Fast

	SATA SSD	vs	NVMe SSD
Interface	SATA		PCIe
Connector	SATA cable		M.2
Speed	up to 550 MB/s		2k-14k+ MB/s
Performance	good		excellent

Quick Quiz

- _____ ?
- _____ ?
- _____ ?

Answers

- _____
- _____

LEARNING GOAL

- ✓ Differentiation, encouraging skills
- ✓ Comparison, mffererecening skills
- ✓ Safe handling skills

Prepared by:
Vishal Sarena

GPUs AND COOLING SYSTEMS



1. WHAT IS A GPU?

DAY 4 MENTION IS REMOVED



Specialized processor for parallel computations & graphics rendering.



USES:



Gaming



Game editing



Artificial Intelligence



Video editing



3D rendering



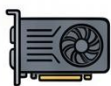
Cryptocurrency mining

TYPES OF GPUs



Integrated GPU (iGPU)

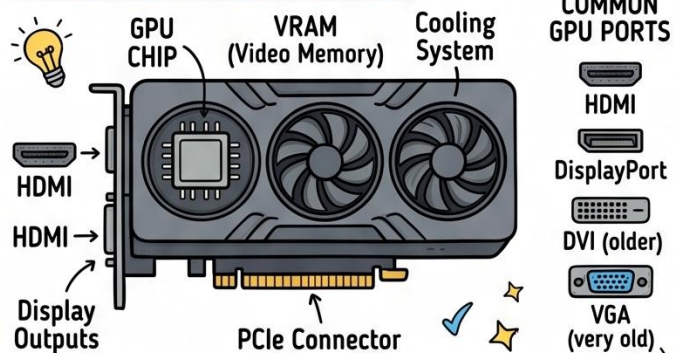
- Built-in
- Intel UHD Graphics
- AMD Radeon (integrated)
- Lower power consumption
- Lower cost
- Lower card maintenance
- Suitable for basic tasks



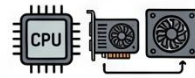
Discrete GPU (Discrete)

- NVIDIA GeForce RTX
- AMD Radeon RX
- Much higher performance
- Dedicated VRAM (video card)
- Better for heavy workloads

2. GPU COMPONENTS



3. COOLING SYSTEMS



Need cooling to operate safely.

4. AIR COOLING

HOW IT WORKS

- 1 Heat moves from CPU/GPU to heatsink
- 2 Heatsink spreads heat
- 3 Fan blows air
- 4 Heat expelled

ADV

- Affordable
- Reliable
- Easy to install
- Low maintenance

DIS

- Can be noisy
- Less efficient

5. LIQUID COOLING

HOW IT WORKS

- 1 Coolant absorbs heat
- 2 Pump circulates coolant
- 3 Radiator releases heat
- 4 Coolant returns

ADV

- Better performance
- Quieter operation
- Attractive appearance

DIS

- More expensive
- More complex
- Potential leak risk (rare)

AIR vs LIQUID



AIR vs LIQUID



PRACTICAL

Activity 1: Identify the GPU

- ✓ Locate card in PCIe x16 slot
- ✓ Expansion card, fans, outputs, power connectors (6/8/12VHPWR)

QUESTIONS

- 1 What are own and erodes?
- 2 What is your work connectors?
- 3 What is sis compenboard?

Activity 3: Identify Case Airflow

- ✓ Find intake & exhaust fans

TYPICAL PATTERN



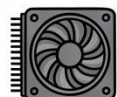
Activity 2: Identify CPU Cooling

- ✓ Look directly above CPU

INDICATORS

Air cooler

Liquid cooler



Air cooler

Liquid cooler

GOOD AIRFLOW



QUICK QUIZ

- 1 What is a questions of ntclent? _____
- 2 What is a GPU's comition? _____
- 3 What is moperatimal questions? _____
- 4 What is cenonove questions? _____

Answers

- 1 Ans ✓
- 2 B, k ✓
- 3 C, sm ✓
- 4 D, wr ✓
- 5 Cra ✓
- 6 D, osr ✓
- 7 VPU ✓
- 8 B/IS ✓

LEARNING GOAL

- ☐ Learnlist checklist of the goal
- ☐ Checklist learn goals
- ☐ Checklist your graphics systems
- ☐ Checklist goal; for learning systems
- ☐ Checklvists of goals

Prepared by:
Vishal Sarena

MOTHERBOARD & CPU

TYPES OF MOTHERBOARDS (FORM FACTORS)



ATX
Standard Size |
Gaming PCs



Micro-ATX
Smaller than ATX |
Budget PCs



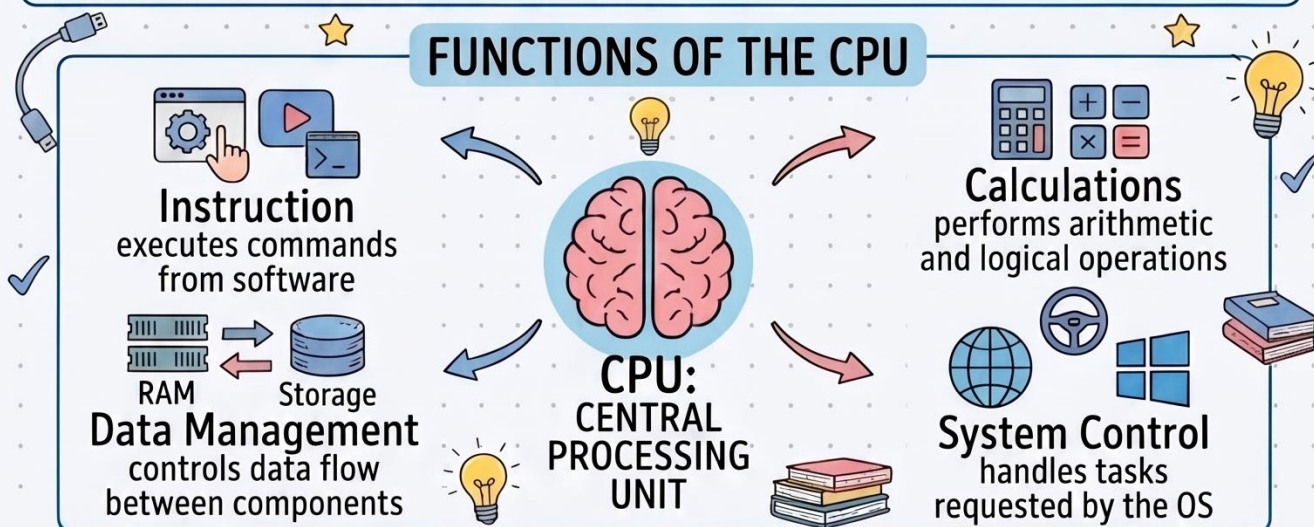
Mini-ITX
Very Compact |
Small Form Factor PCs



E-ATX
Larger than ATX |
Workstations



FUNCTIONS OF THE CPU

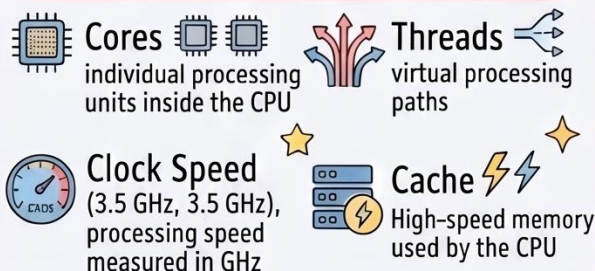


CPU SOCKET COMPATIBILITY

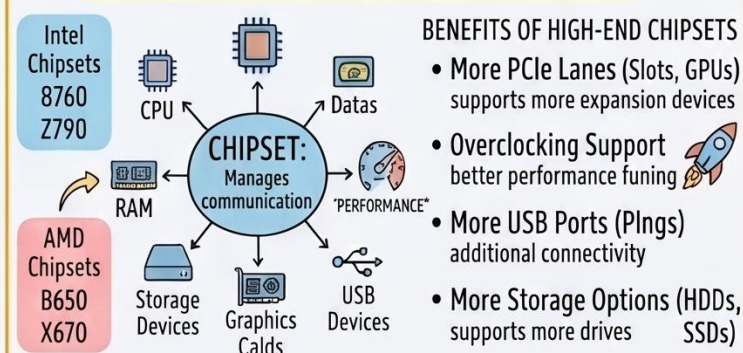
SOCKET TYPE	SUPPORTED PROCESSORS
Intel LGA 1700	12th, 13th, 14th Gen Intel CPUs
AMD AM4	Ryzen 1000-5000 Series
AMD AM5	Ryzen 7000 Series and Later

★ NOTE: A processor cannot be installed in a different socket type!

IMPORTANT CPU SPECIFICATIONS



CHIPSET FUNCTIONS & EXAMPLES



QUICK TECH COMMANDS

- msinfo32 (system info)
- diskmgmt.msc (disks)
- Shrink Volume: Select Drive -> Right click -> shrink volume -> select size -> OK
- Extend Volume: Select Drive -> Right click -> shrink volume -> select size -> OK [Select Old Drive]
- NTFS, FAT32: File Systems
- cmd -> properties -> select font size
- color OA. List of colors (0 = Black, 2 = Green, 7 = White...)
- cls: Clean terminal



YOU GOT THIS! VISHAL SARENA

ESSENTIAL WINDOWS KEYBOARD SHORTCUTS & CMD COMMANDS



1 USEFUL KEYBOARD SHORTCUTS

SHORTCUT	FUNCTION
Windows + E	Open File Explorer
Windows + R	Open Run Dialog Box
Windows + D	Show Desktop
Windows + L	Lock Computer
Windows + Tab	Task View
Alt + Tab	Switch Applications
Ctrl+Shift+Esc	Open Task Manager Directly
Ctrl + A	Select All
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + X	Cut
Ctrl + Z	Undo
Ctrl + Y	Redo
F2	Rename File
F5	Refresh
Windows + Shift + S	Screenshot Tool



SHORTCUT	FUNCTION
• cmd	Command Prompt
• msconfig	System Configuration
• taskmgr	Task Manager
• control	Control Panel
• appwiz.cpl	Installed Programs (Control panel or applications)
• devmgmt.msc	Running Processes
• alt+cpl	Clear All



2 RUN COMMANDS (WIN + R)

Press Windows + R and try these commands.

- cmd
 - msconfig
 - taskmgr
 - control
 - appwiz.cpl
 - devmgmt.msc
 - diskmgmt.msc
 - services.msc
 - ncpa.cpl
 - calc
 - notepad
 - dxdiag
 - cleanmgr
- Command Prompt
 - System Configuration
 - Task Manager
 - Control Panel
 - Installed Programs
 - Device Manager
 - Disk Management
 - Windows Services
 - Network Connections
 - Calculator
 - Notepad
 - DirectX Information
 - Disk Cleanup

Activity: Students open 5 different tools using Run commands.

3 BASIC CMD COMMANDS



Open Command Prompt.

- hostname
 - whoami
 - ipconfig
 - ipconfig /all
 - systeminfo
 - tasklist
 - cls
 - date
 - time
 - shutdown /s /t 60
 - shutdown /a
 - tree
 - dir
- Shows PC Name
 - Shows Current User
 - Shows IP Address
 - Detailed Network Info
 - Complete System Info
 - Running Processes
 - Clear Screen
 - Show Current Date
 - Show Current Time
 - Shutdown after 60 Secs
 - Cancel Shutdown
 - Display Folder Structure
 - Show Files and Folders

LEARN & GROW!
STAY CURIOUS!



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